

AUGUST/RESIT EXAMINATIONS 2023/2024

MODULE: CA266 - Probability & Statistics

PROGRAMME(S):

COMSCI	BSc in Computer Science
ECSAO	Study Abroad (Engineering & Computing)
SHSAO	Study Abroad (Science & Health)

YEAR OF STUDY: 2,0

EXAMINER(S):

Dr. Graham Healy	(Internal)	(Ext:7364)
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TIME ALLOWED: 2 Hours

INSTRUCTIONS: Answer 4 questions. All questions carry equal marks.

PLEASE DO NOT TURN OVER THIS PAGE UNTIL YOU ARE INSTRUCTED TO DO SO.

The use of programmable or text storing calculators is expressly forbidden.

Please note that where a candidate answers more than the required number of questions, the examiner will mark all questions attempted and then select the highest scoring ones.

There are no additional requirements for this paper.

QUESTION 1**[TOTAL MARKS: 25]**

For each of these questions **explain your answer** i.e., show your reasoning.

Q 1(a)**[12 Marks]**

Suppose you have 2 boxes each of which contains six balls. The first box contains a red ball, a blue ball, a green ball, a purple ball, a white ball, and a yellow ball. The second box contains 2 red balls, 2 blue balls and 2 purple balls.

If you draw a ball at random from the first box, and then again at random from the second box, what is the probability that:

- I. the first ball and the second ball are red?
- II. the second ball is red given that the first ball was red?
- III. the first ball or the second ball or both are purple?
- IV. both of the balls are not the same colour?

Q 1(b)**[13 Marks]**

A batch of 40 components contains 30 that are defective. Three are selected at random **without replacement**.

- I. What is the probability that the second component is defective given that the first was not defective?
- II. What is the probability that all three components are defective?
- III. What is the probability that at least one of the three components are defective?
- IV. What is the probability exactly two of the components are defective?
- V. For any of the questions in Q1(b) I-IV, explain how your answer would change if sampling with replacement was used.

[End of Question 1]

QUESTION 2**[TOTAL MARKS: 25]**

For each of these questions **explain your answer** i.e., show your reasoning.

A company manufactures the same type of car at three different locations. After cars are manufactured they are sent to a warehouse in Belgium for testing to find defects.

The following table shows the breakdown of percentages of defects from each of the locations e.g. 10% of the cars manufactured in Ireland has 0 defects.

		Defects (number of defects)			
		0	1	2	3
Location	Ireland	10%	10%	5%	15%
	France	5%	0%	10%	5%
	Spain	10%	20%	10%	0%

Q 2(a)**[10 Marks]**

Given values for the following probabilities, if a car was selected at random from the warehouse in Belgium:

- i. $P(\text{Location} = \text{Ireland})$
- ii. $P(\text{Defects} < 2)$
- iii. $P(\text{Defects} = 2)$
- iv. $P(\text{Defects} = 2 \text{ AND Location} = \text{Spain})$
- v. $P(\text{Defects} = 0 \mid \text{Location} = \text{Ireland})$
- vi. $P(\text{Location} = \text{Spain} \mid \text{Defects} = 1)$
- vii. $P(\text{Location} = \text{France OR Defects} = 2)$

Q 2(b)**[8 Marks]**

What are the mean and variance of the number of defects for the cars in the Belgian warehouse?

Q 2(c)**[7 Marks]**

You are told there are 1000 cars in the warehouse. What is the probability that if you selected two cars at random, that both were manufactured at the same location?

[End of Question 2]

QUESTION 3**[TOTAL MARKS: 25]**

In a university department, 20% of people use Word, 20% use LaTeX, and 60% use HTML. From experience it is known that:

- 30% of the Word documents contain errors.
- 10% of the LaTeX documents contain errors.
- 10% of the HTML documents contain errors.

For each of these questions **explain your answer** i.e., show your reasoning.

Q 3(a)**[3 Marks]**

What is the probability that a document does **not** contain errors, given that it is written in HTML?

Q 3(b)**[6 Marks]**

If a document is chosen at random, what is the probability that it will contain errors?

Q 3(c)**[9 Marks]**

A document is chosen at random and is found to contain no errors. Is it most probable the document was written in Word, LaTeX, or HTML? **Explain your answer.**

Q 3(d)**[3 Marks]**

A worker is inspecting documents, one by one, until they find a LaTeX or Word document. What is the probability that the first LaTeX or Word document they find will be the 6th document they inspect?

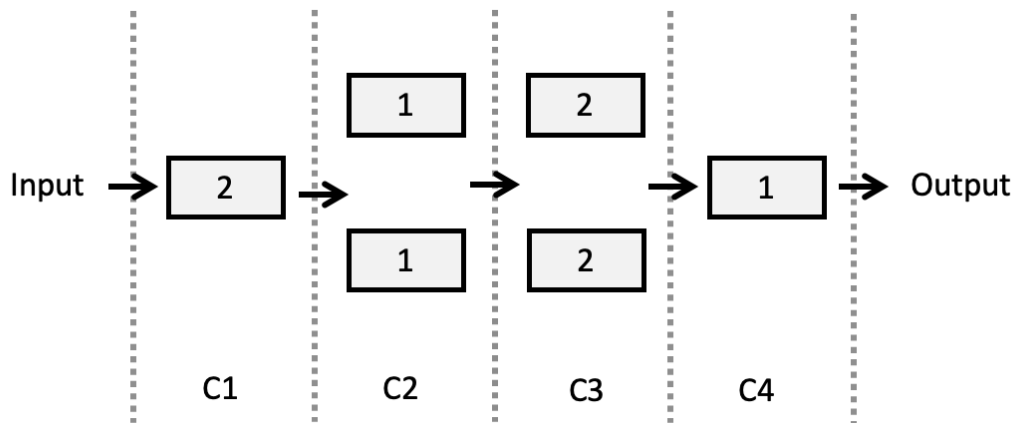
Q 3(e)**[4 Marks]**

A worker selects 10 random documents from a folder than contains 20 Word documents, 20 LaTeX documents, and 60 HTML documents. What is the probability that there will be exactly 5 LaTeX documents in their selection?

[End of Question 3]

QUESTION 4**[TOTAL MARKS: 25]**

The following system consists of a number of components in series and in parallel. The number in the box is the mean number of failures for that type of component in a period of 1 year. There are four (4) types of components in this system: C1, C2, C3 and C4. Assume the probability of the failures follow a Poisson distribution.



For each of the questions below **explain your answer** i.e., show your reasoning.

Q 4(a) **[4 Marks]**

For each component, give the probability that it will still function at the end of the 1-year period.

Q 4(b) **[4 Marks]**

For each component, give the probability that it will fail 2 times in a 1-year period.

Q 4(c) **[7 Marks]**

What is the probability that the whole system will still function at the end of the 1-year period?

Q 4(d) **[6 Marks]**

How long would component C1 need to be in operation for there to be a .99 probability that a failure of this component will occur?

Q 4(e) **[4 Marks]**

An engineer is studying component C1. Given that the component has not failed in the past 6 months, what is the probability that it will fail in the following 6 months?

[End of Question 4]

QUESTION 5**[TOTAL MARKS: 25]**

Answer any 2 of the following questions i.e. of Q5(a), Q5(b) or Q5(c). There are equal marks for both questions.

For each of these questions **explain your answer** i.e., show your reasoning.

Q 5(a)**[12.5 Marks]**

A person is tossing a fair coin repeatedly, where each coin toss results in a head or tail (with equal probability). Give values for the following probabilities:

- I. If the previous 5 coin tosses were all heads, what is the probability the next 5 coin tosses will be tails?
- II. If the person tosses the coin 10 times, what is the probability 7 of these tosses will be heads?
- III. If the person tosses the coin 100 times, what is the expected variance of the number of heads?

Q 5(b)**[12.5 Marks]**

Person A and person B have taken part in a competition every week for the past number of years. We are told that the average score for both contestants is 50. We are told the standard deviation of the scores for Person A is 3. We are not told the standard deviation of the scores for person B. You can assume the scores for both person A and person B are normally distributed.

- I. What is the probability that person A scored above 51.5?
- II. What is the probability that person A scored above 53, if we are told their score is above 47?
- III. We are told there is a 0.9332 probability that person B scored below 53. What is the standard deviation for person B?

Q 5(c)**[12.5 Marks]**

We throw two 6-faced dice, let the total score be X .

- I. What are the probabilities for each value of X ?
- II. If we are told that the first throw was greater than 2, what is the probability that $X > 3$?
- III. If we are told that **both** throws of the dice were an odd number, what is the probability that $X < 7$?

[End of Question 5]

APPENDICES

Area under the standard normal curve from $-\infty$ to z



z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6338	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990

${}^nC_k = \frac{{}^nP_k}{k!} = \frac{n!}{k!(n-k)!}$	
${}^nP_k = n(n-1)\dots(n-k+1) = \frac{n!}{(n-k)!}$	
$P(E_1 \cup E_2) = P(E_1) + P(E_2) - P(E_1 \cap E_2)$	
$P(A B) = \frac{P(A)P(B A)}{P(B)}$	$P(A \cap B) = \frac{P(A)P(B A)}{P(B)P(A B)}$
$V(X) = \sum_x (x - \mu_x)^2 p(x)$	$E(X) = \sum_x xp(x)$
$P(X = x) = \binom{n}{x} p^x q^{n-x}$	$P(X \leq x) = 1 - q^x$ $P(X = x) = q^{x-1} p$
$P(X = x) = \frac{\binom{M}{x} \binom{L}{n-x}}{\binom{N}{n}}$	$P(X = x) = e^{-\lambda} \frac{\lambda^x}{x!}$

[END OF APPENDICES]

[END OF EXAM]